

FRONTEND DEVELOPMENT

5.1 Introduction

The frontend of the Stock Trading Application is developed using React.js with Vite. It provides a fast, responsive, and interactive user interface. React components improve code reusability while Vite offers fast development and optimized builds. The frontend communicates with the backend to perform registration, login, portfolio management, and stock trading operations.

5.2 Frontend Project Structure

The project contains folders such as assets, components, pages, and services. Components store reusable UI elements, pages store complete screens, services manage API calls, and assets contain images and icons.

5.3 Client Setup

The project is created using Vite. Dependencies are installed using npm. React Router DOM manages navigation and Axios handles API communication. The application is started using `npm run dev`.

5.4 Component-Based Development

React follows a component-based architecture. Independent components such as Navbar, Login, Register, Dashboard, Portfolio, and Stock Cards simplify development and maintenance.

5.5 Application Routing

React Router enables navigation between Home, Login, Register, Dashboard, Portfolio, and Trading pages without refreshing the browser.

5.6 User Interface Development

HTML, CSS, and React components create a responsive interface with consistent styling and improved user experience.

5.7 API Integration

Axios sends HTTP requests to the Express backend. The backend processes requests and interacts with MongoDB before returning JSON responses.

5.8 Frontend Workflow

User → React Components → React Router → Axios → Express Backend → MongoDB → Response → Updated User Interface.

5.9 Advantages

Reusable components, faster rendering, modular structure, responsive design, easy maintenance, scalability, and efficient API communication.

5.10 Summary

The frontend architecture provides a scalable and user-friendly interface that integrates seamlessly with the MERN stack backend.

File

Edit

Selection

View

Go

Run

Terminal

Help

stock Trading

JS userController.js

JS JS

JS userRoutes.js

JS authMiddleware.js

JS db.js

JS server.js

STOCK TRADING

client

src

App.css

App.jsx

index.css

main.jsx

.gitignore

eslint.config.js

index.html

package-lock.json

package.json

README.md

vite.config.js

server

config

db.js

controllers

userController.js

middleware

authMiddleware.js

models

User.js

node_modules

routes

userRoutes.js

uploads

utils

.env

package-lock.json

package.json

server.js

JS JS

OUTLINE

TIMELINE

JS JS > App

1 import { BrowserRouter, Routes, Route } from "react-router-dom";

2

3 import Navbar from "../components/Navbar";

4

5 import Home from "../pages/Home";

6 import Login from "../pages/Login";

7 import Register from "../pages/Register";

8 import Dashboard from "../pages/Dashboard";

9 import Portfolio from "../pages/Portfolio";

10

11 function App() {

12

13 return (

14

15 <BrowserRouter>

16

17 <Navbar />

18

19 <Routes>

20

21 <Route path="/" element={<Home />} />

22

23 <Route path="/login" element={<Login />} />

24

25 <Route path="/register" element={<Register />} />

26

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\AKHILA\OneDrive\stock Trading\server> npm run dev

> server@1.0.0 dev

> nodemon server.js

[nodemon] 3.1.14

[nodemon] to restart at any time, enter `rs`

[nodemon] watching path(s): *.*

[nodemon] watching extensions: js,mjs,cjs,json

[nodemon] starting `node server.js`

◇ injected env (0) from .env // tip: = custom filepath { path: '/custom/path/.env' }

Server is running on port 5000

[nodemon] clean exit - waiting for changes before restart

powerShell

node

powerShell

node

node

powerShe...

powerShe...

node server

node server

Ln 27, Col 60

Spaces: 4

UTF-8

CRLF

JavaScript

Go Live

Prettier

```
+ v ... x
powershell
└─ node
  └─ powershell
    └─ node
      └─ node
        └─ powershell...
          └─ powershell...
            └─ node server
              └─ node server
```

